Exercises

(1) If an input **x** = (1, 4) is presented to a network with two output units with weight vectors $\mathbf{w}_1$ = (2, 3) and $\mathbf{w}_2$ = (3, 2), which output unit will be the winner and why? Draw the vectors in input space and calculate the inner product of **x** with $\mathbf{w}_1$ and $\mathbf{w}_2$.

(2) Competitive learning moves weight vectors in input space so that the weight vectors of different output units move to the centres of clusters input space. Why is it beneficial for the weight vectors to move to the centres of the clusters?

(3) Consider a Kohonen network with a 2D array of 5 x 5 output units and two inputs x_1 and x_2 that are chosen from a flat random distribution between 2 and 8. Draw the final state of the network both in input and output space.

(4) Most self-organising maps map an N-dimensional input space onto a 2-dimensional (physical) output space. Give examples of 2D – 2D, 1D – 2D and 2D – 1D mappings.